

Homework Five, Part 1

GSE 544 — Spring 2026

This is the first of the two computational problems that make up Homework Five. It is meant to be done in a Jupyter notebook: write the code yourself (liberally using the code in the github repo for the class as starting point is fine), produce the requested tables and figures, and answer the written questions in markdown cells. The deliverable is a url from your github repo with the corresponding Jupyter notebook. The data file referenced below is available on the github repo for the class, in `06_Portfolio_optimization_and_copulas/` (`vmls_portfolio_returns.csv`).

Fitting marginals and copulas to the VMLS portfolio (50 points)

In the Topic-6 lecture notebook we fit Student- t marginals to each of the 19 risky assets, took PITs, and built a **Gaussian** copula on the joint distribution. Gaussian was forced by 19-dimensional intractability, but it has a real cost: a Gaussian copula has *zero* asymptotic tail dependence, so even after the marginals capture each asset's fat tail, the joint model loses the co-movement in the tails. Here you make that cost visible on a tractable **four-asset** subset.

Use the training set (first 2000 rows of `vmls_portfolio_returns.csv`), 19 risky assets indexed 1–19. Work with the four-asset sample **stocks 1, 2, 3, 10** (0-indexed `[0, 1, 2, 9]`) for all plots and copula fits.

Part (a) [6 points] — Student- t marginals

Fit a Student- t distribution to each of the 19 assets and store the `(df, loc, scale)` parameters. Print the summary for stocks 1, 2, 3, 10. Confirm in one sentence that all four have $\nu < 5$ (very fat tails), and note which of the four is fattest-tailed and which is lightest.

Part (b) [12 points] — PITs and the empirical scatter matrix

Apply the probability integral transform $u = F(r)$ to every column using its fitted marginal CDF, and clip the result to $[10^{-6}, 1 - 10^{-6}]$ (so that Φ^{-1} and t^{-1} stay finite downstream). Then draw the 4×4 **empirical PIT scatter matrix** for the four-asset sample: diagonal panels = marginal-uniformity histograms, off-diagonal panels = pairwise scatters of (u_j, u_k) . In two or three sentences, describe what you see — are the diagonal histograms approximately uniform, and where in the off-diagonal panels does the dependence concentrate? This empirical scatter is the benchmark any fitted copula must reproduce.

Part (c) [8 points] — Gaussian copula

Fit the Gaussian copula on the four-asset subset. That single matrix is the *entire* dependence structure of a Gaussian copula.

Part (d) [12 points] — t -copula

Fit a t -copula on the same subset. How does the degrees of freedom from the copula $\hat{\nu}$ compare to the marginal dfs?

Part (e) [12 points] — Simulate and compare

Draw $n = 2000$ uniform-margin observations from **each** fitted copula (matching the training size) and plot their PIT scatter matrices in the same 4×4 layout as part (b). Put the empirical, Gaussian-simulated, and t -simulated scatter matrices side by side and compare the corner (bottom-left / top-right) mass across the three. In a few sentences, state which copula reproduces the empirical corners better and why this is the canonical reason finance prefers the t -copula.